ω Class 4:

1. Find the Fourier sine transform of the function $f(t) = \begin{cases} t, & t < \frac{\pi}{2} \\ 0, & t > \frac{\pi}{2} \end{cases}$

Ans: $\frac{1}{\omega^2} sin\left(\frac{\pi\omega}{2}\right) - \frac{\pi}{2\omega} cos\left(\frac{\pi\omega}{2}\right)$

2. Find the Fourier sine transform of $2e^{-5t} + 5e^{-2t}$. Ans: $\omega\left(\frac{2}{\omega^2+25} + \frac{5}{\omega^2+4}\right)$.

3. Find the Fourier sine transform of $f(t) = e^{-at}, a > 0$. Show that $\int_0^\infty \frac{x\sin kx}{a^2+x^2}\,dx = \frac{\pi}{2}e^{-ax}, (k > 0)$ Ans: $\frac{\omega}{\omega^2+a^2}$

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